Task 7: Software Cost Estimation and Off-the-Shelf Evaluation

Project Title: Final Year Project Management System (FYPMS)

Project Description: A centralized web-based system to manage the entire lifecycle of final year university projects. Key functionalities include project proposal submission, supervisor allocation, milestone tracking, and document submission, plagiarism checking integration, communication tools, and final grading.

1. Identification of Off-the-Shelf Software Options and Justification

Instead of building a system from scratch, the university can consider adopting and customizing an existing off-the-shelf (OTS) software package. This approach can significantly reduce development time, cost, and risk.

We have identified three primary categories of OTS solutions:

Option A: Project Management Platforms (e.g., ClickUp, Asana)

- ✓ Justification for Use: These are highly mature, cloud-based platforms with robust task management, timeline (Gantt chart), collaboration, and document sharing features. They are designed to manage complex projects and could be repurposed for academic projects. Their primary advantage is a low initial cost and instant deployment.
- ✓ Suitability for FYPMS: Medium. They lack native academic-specific features like formal proposal templates, supervisor allocation algorithms, integrated plagiarism checks, and a structured grading workflow. These would require significant "workaround" configurations.
- ✓ Option B: Learning Management Systems (LMS) with Advanced Features (e.g., Moodle, Canvas)
- ✓ Justification for Use: The University already has an LMS in place (e.g., Moodle). Modern LMS platforms offer plugins and modules for managing assignments, groups, and rubrics. Leveraging the existing system reduces cost, and users (students and staff) are already familiar with the environment.

- ✓ Suitability for FYPMS: High. With custom plugin development, an LMS can be tailored to handle the FYP lifecycle. It natively supports submission portals, grading rubrics, and user roles. However, deep customization is still required.
- ✓ Option C: Low-Code/No-Code Platform (e.g., Microsoft Power Apps, Bubble.io)
- ✓ Justification for Use: These platforms allow for the rapid development of custom business applications with minimal hand-coding. They offer a middle ground between a rigid OTS product and a fully custom-built system, enabling the creation of a tailor-made FYPMS without the full cost of a development team.
- ✓ Suitability for FYPMS: High. A Power App, for instance, could be built to integrate seamlessly with the universities existing Microsoft 365 ecosystem (Teams, SharePoint, OneDrive), providing a highly customized and integrated solution at a fraction of the cost of custom software.

2. Cost Estimation Table

The following table provides a high-level cost comparison between the OTS/Customization approach and a traditional custom software development approach. Costs are estimated in US Dollars for a 3-year lifecycle.

Assumptions:

- User Base: 500 students, 50 supervisors/staff.
- Development is outsourced to a third-party vendor at an average rate of $50/hour.
- "Low Customization" refers to configuration and basic workflow setup. "High Customization" refers to developing new plugins, modules, or complex automations.

Cost Component	Option A: Click Up/Asana (OTS + Config)	Option B: Moodle (OTS + Custom Plugin)	Option C: MS Power Apps (Low-Code Platform)	Option D: Custom-Built System (From Scratch)
Software License (One-Time)				
Setup & Configuration (40 hours)	**$2,000**	**$2,000**	**$2,000**	
Custom Development (Plugin/App)		$15,000 (300 hours)	$10,000 (200 hours)	$75,000 (1500 hours)
Subtotal (One-Time)	**$2,000**	**$17,000**	**$12,000**	**$75,000**
Recurring Costs (Annual)			$4,950 (Power Apps Premium)	
User Licenses (550 users)	$8,250		$4,950 (Power Apps Premium)	
Cloud Hosting & Maintenance		$2,000	Included in license	$5,000
Technical Support (20 hours/yr)	$1,000	$1,000	$1,000	$2,000
Annual Recurring Subtotal	**$9,250**	**$3,000**	**$5,950**	**$7,000**

Total 3-Year Cost	**$2,000 + ($9,250 x 3) = $29,750**	**$17,000 + ($3,000 x 3) = $26,000**	**$12,000 + ($5,950 x 3) = $29,850**	**$75,000 + ($7,000 x 3) = $96,000**

3. Recommendation Report for Software Package

To: University IT Steering Committee

From: [Your Name/Team Name], Project Analyst

Date: October 26, 2023

Subject: Recommended Software Solution for the Final Year Project Management System (FYPMS)

1. Executive Summary

After a thorough analysis of off-the-shelf (OTS) options and a custom development approach, Option C: Microsoft Power Apps is the recommended solution for the FYPMS. It provides the optimal balance of cost-effectiveness, customization capability, and strategic alignment with the university's existing IT infrastructure. This approach is projected to save the university over $66,000 compared to a custom-built system over three years while delivering a purpose-built application.

2. Evaluation of Options

- Option A (ClickUp/Asana): While initially appealing due to its low configuration cost, the high recurring license fees and fundamental misalignment with academic processes make it a less suitable long-term solution. It would force the university to adapt its processes to the software, not the other way around.
- Option B (Moodle + Custom Plugin): This is a strong contender. It leverages existing systems and has moderate costs. However, development is dependent on Moodle's specific framework, which can be less flexible and may result in higher long-term maintenance costs compared to a more modern low-code platform.

- Option C (MS Power Apps): This option strikes the perfect balance. It allows us to build a fully custom FYPMS that fits our exact workflow without the exorbitant cost of traditional development. Its deep integration with Microsoft 365 (which the university already pays for and uses) means seamless single sign-on, data storage in SharePoint, and collaboration via Teams.
- Option D (Custom-Built): This option offers maximum flexibility and control but at a prohibitively high cost and with the longest development timeline. The risks associated with scope creep, project delays, and finding a skilled development team are significant.

3. Final Recommendation: Microsoft Power Apps

We strongly recommend proceeding with the development of the FYPMS using the Microsoft Power Platform (Power Apps)

Justification:

- Cost-Effectiveness: At an estimated 3-year cost of $29,850, it is 69% cheaper than a custom solution and competitive with other OTS options while offering superior customization.
- Customization and Fit: Power Apps will allow us to build a system that mirrors the university's unique FYP workflow, from proposal to grading, ensuring high user adoption and satisfaction.
- Strategic Integration: The solution will natively integrate with the university's Microsoft 365 tenant, leveraging existing security, user accounts (Azure AD), and collaboration tools (Teams, SharePoint), reducing IT overhead.
- Scalability and Maintainability: The platform is managed by Microsoft, ensuring high availability and security. Future changes and enhancements can be made relatively quickly and inexpensively compared to hard-coded software.
- Reduced Risk: The low-code approach is faster to prototype and deploy, allowing for iterative feedback from stakeholders and reducing the risk of building a system that does not meet user needs.

4. Next Steps

1. Form a Project Team: Assemble a small team comprising an IT staff member (Power Platform developer), a representative from the academic registry, and a lead supervisor.

2. Develop a Prototype: Use the allocated budget to build a minimum viable product (MVP) focusing on the core workflow of project proposal and allocation.

3. Pilot and Iterate: Run a pilot program with one academic department, gather feedback, and refine the application before a full university-wide rollout.

We are confident that a Power Apps-based FYPMS will provide an excellent return on investment and serve the university's needs effectively for years to come.